Changes in Adolescent Alcohol Use

Group A

2026-03-07

Data Summary

```
alc <- read.table("alcohol.txt", header=TRUE) # loading the dataset
alc <- alc %>%
  mutate(coa = factor(coa, levels=c("0", "1")), # convert parental alcoholism to a factor
                      male=factor(male, levels=c("0", "1"))) # convert gender to a factor

head(alc)
```

```
##   id age coa male   alcuse      peer
## 1  1  14   1    0 1.732051 1.2649111
## 2  1  15   1    0 2.000000 1.2649111
## 3  1  16   1    0 2.000000 1.2649111
## 4  2  14   1    1 0.000000 0.8944272
## 5  2  15   1    1 0.000000 0.8944272
## 6  2  16   1    1 1.000000 0.8944272
```

```
alc_base <- alc %>% # extract baseline observations (age 14)
  group_by(id) %>%
  slice(1) %>%
  ungroup()

summary_table <- bind_rows(

  tibble(
    Characteristic = "Alcohol use",
    Value = ""
  ),

  # summarizing alcohol use for each age
  bind_rows(lapply(sort(unique(alc$age)), function(a) {
    dat <- alc %>% filter(age == a)
    tibble(
      Characteristic = c(
        paste0("\\hspace{1em}Age ", a),
        "\\hspace{2em}Mean (SD)",
        "\\hspace{2em}Median",
        "\\hspace{2em}Range"
      ),
      Value = c(
```

```r
      "",
      sprintf("%.2f (%.2f)", mean(dat$alcuse, na.rm = TRUE), sd(dat$alcuse, na.rm = TRUE)),
      sprintf("%.2f", median(dat$alcuse, na.rm = TRUE)),
      sprintf("%.2f--%.2f", min(dat$alcuse, na.rm = TRUE), max(dat$alcuse, na.rm = TRUE))
    )
  )
}),

tibble( # creating a tibble to summarize peer alcohol use
  Characteristic = c(
    "Peer alcohol use",
    "\\hspace{1em}Mean (SD)",
    "\\hspace{1em}Median",
    "\\hspace{1em}Range"
  ),
  Value = c(
    "",
    sprintf("%.2f (%.2f)", mean(alc_base$peer, na.rm = TRUE), sd(alc_base$peer, na.rm = TRUE)),
    sprintf("%.2f", median(alc_base$peer, na.rm = TRUE)),
    sprintf("%.2f--%.2f", min(alc_base$peer, na.rm = TRUE), max(alc_base$peer, na.rm = TRUE))
  )
),

# summarize the proportion of each gender
alc_base %>%
  mutate(
    male_label = case_when(
      male == "0" ~ "Female",
      male == "1" ~ "Male",
      TRUE ~ as.character(male)
    )
  ) %>%
  count(male_label) %>%
  mutate(
    Characteristic = paste0("\\hspace{1em}", male_label),
    Value = paste0(n, " (", round(100 * n / sum(n), 1), "\\%)")
  ) %>%
  select(Characteristic, Value) %>%
  bind_rows(tibble(Characteristic = "Gender", Value = ""), .),

# summarizing the proportion of COA = 0 vs 1
alc_base %>%
  mutate(
    coa_label = case_when(
      coa == "0" ~ "No",
      coa == "1" ~ "Yes",
      TRUE ~ as.character(coa)
    )
  ) %>%
  count(coa_label) %>%
  mutate(
    Characteristic = paste0("\\hspace{1em}", coa_label),
    Value = paste0(n, " (", round(100 * n / sum(n), 1), "\\%)")
```

```r
    ) %>%
    select(Characteristic, Value) %>%
    bind_rows(tibble(Characteristic = "Parental alcoholism", Value = ""), .)
)

kable(
  summary_table,
  format = "latex",
  booktabs = TRUE,
  caption = "Summary of study variables",
  col.names = c("Variables", "Value"),
  align = "ll",
  linesep = "",
  escape = FALSE
) %>%
  kable_styling(
    latex_options = "hold_position",
    font_size = 10
  )
```

Table 1: Summary of study variables

Variables	Value
Alcohol use	
Age 14	
Mean (SD)	0.63 (0.94)
Median	0.00
Range	0.00–3.32
Age 15	
Mean (SD)	0.96 (1.05)
Median	1.00
Range	0.00–3.61
Age 16	
Mean (SD)	1.17 (1.14)
Median	1.00
Range	0.00–3.46
Peer alcohol use	
Mean (SD)	1.02 (0.73)
Median	0.89
Range	0.00–2.53
Gender	
Female	40 (48.8%)
Male	42 (51.2%)
Parental alcoholism	
No	45 (54.9%)
Yes	37 (45.1%)

Missing data

```
missing_table <- alc %>% # summarize missing data by age
  group_by(age) %>%
  summarise(
    "Number of observations" = n(),
    "Alcohol use" = sum(is.na(alcuse)),
    "Peer alcohol use" = sum(is.na(peer)),
    "Parental alcoholism" = sum(is.na(coa)),
    "Gender" = sum(is.na(male))
  )

paper_kable(missing_table, "Missing data summary by age")
```

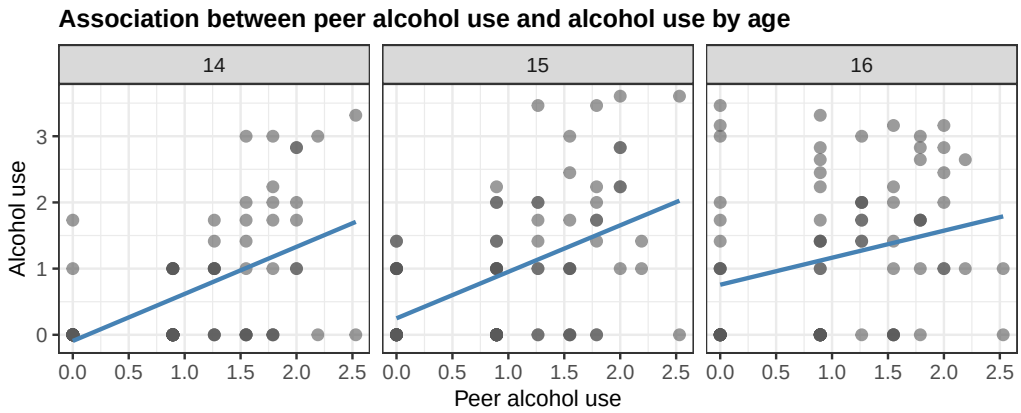
Table 2: Missing data summary by age

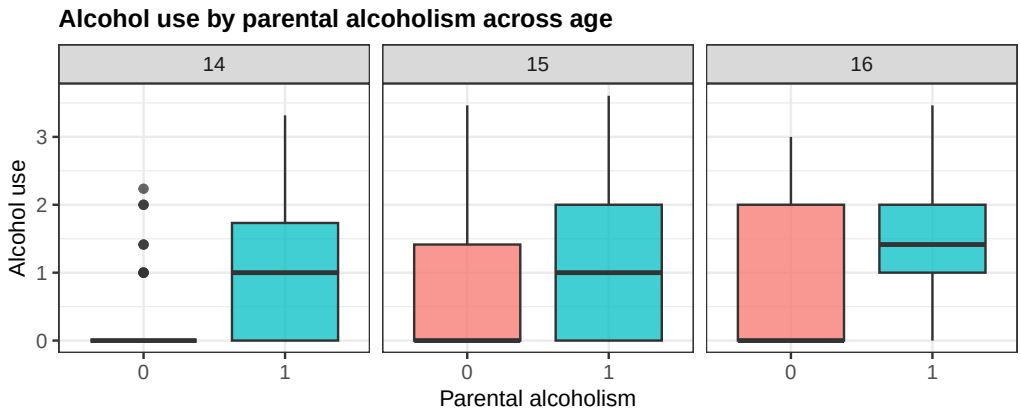
age	Number of observations	Alcohol use	Peer alcohol use	Parental alcoholism	Gender
14	82	0	0	0	0
15	82	0	0	0	0
16	82	0	0	0	0

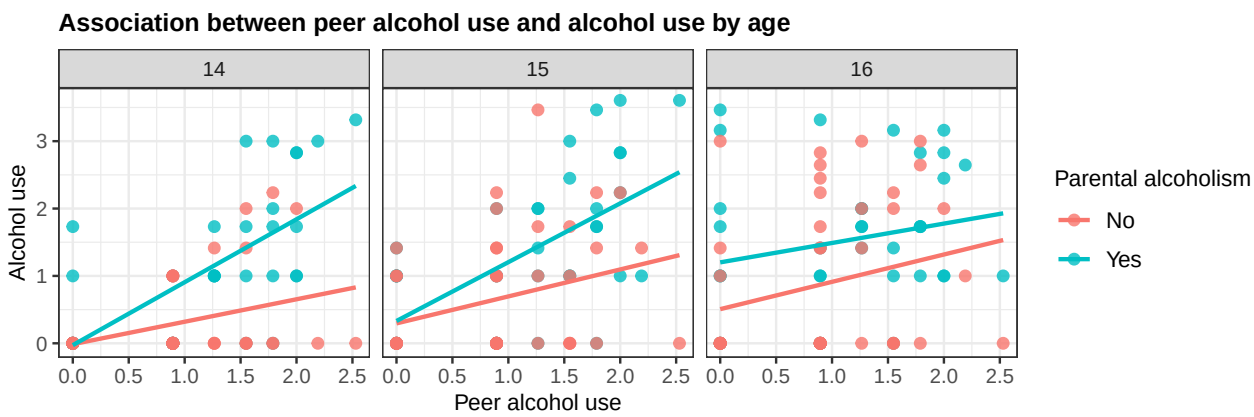
Exploratory plots

```
# creating plots to visualize relations
p_box_age <- ggplot(alc, aes(x = coa, y = alcuse, fill = coa)) +
  geom_boxplot(alpha = 0.75, show.legend = FALSE) +
  facet_wrap(~ age) +
  scale_fill_manual(values = coa_cols) +
  labs(
    title = "Alcohol use by parental alcoholism across age",
    x = "Parental alcoholism",
    y = "Alcohol use"
  ) +
  base_plot_theme

p_peer_age <- ggplot(alc, aes(x = peer, y = alcuse)) +
  geom_point(alpha = 0.6, size = 2, color = "grey35") +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.9, color = "steelblue") +
  facet_wrap(~ age) +
  labs(
    title = "Association between peer alcohol use and alcohol use by age",
    x = "Peer alcohol use",
    y = "Alcohol use"
  ) +
  base_plot_theme

p_peer_age_coa <- ggplot(alc, aes(x = peer, y = alcuse, color = coa)) +
  geom_point(alpha = 0.8, size = 2) +
```

```r
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.9) +
  facet_wrap(~ age) +
  labs(
    title = "Association between peer alcohol use and alcohol use by age",
    x = "Peer alcohol use",
    y = "Alcohol use",
    color = "Parental alcoholism"
  ) +
  scale_color_manual(
    values = coa_cols,
    labels = c("No", "Yes")
  ) +
  base_plot_theme

# grouping the tables onto the same page
eda_plots <- (
  p_peer_age +
  p_box_age +
  p_peer_age_coa
) +
  plot_layout(ncol = 1)

eda_plots
```



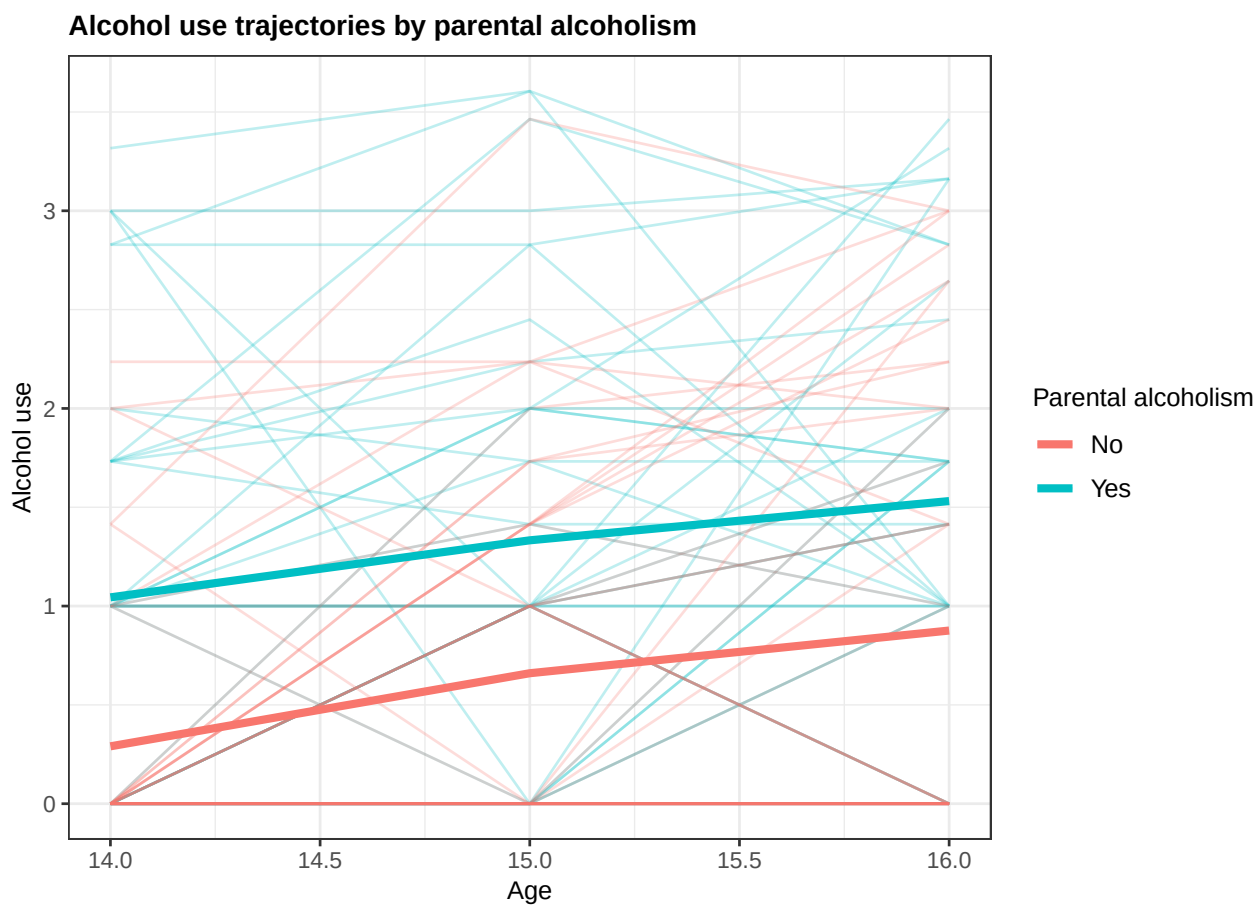




Individual trajectories with mean trend

```
# plotting the trajectory plot
ggplot(alc, aes(x = age, y = alcuse, group = id, color = coa)) +
  geom_line(alpha = 0.25) +
  stat_summary(
    aes(group = coa, color = coa),
    fun = mean,
    geom = "line",
    linewidth = 1.5
  ) +
  scale_color_manual(values = coa_cols, labels = c("No", "Yes"),
```

```
                      name = "Parental alcoholism") +
  labs(
    title = "Alcohol use trajectories by parental alcoholism",
    x = "Age",
    y = "Alcohol use"
  ) +
  base_plot_theme
```



Model comparison

```
m1 <- lme( # random intercept + ind errors model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  data = alc,
  method = "REML"
)

m2 <- lme( # random intercept and slopes + ind errors model
  alcuse ~ age * coa * peer * male,
  random = ~ age | id,
  data = alc,
  method = "REML"
```

```r
)

m3 <- lme( # random intercept + AR1 errors model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "REML"
)

m4 <- lme( # random intercept + exp errors model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  correlation = corExp(form = ~ age | id),
  data = alc,
  method = "REML"
)

m5 <- lme( # random intercept + heteroske ind errors (age) model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  weights = varIdent(form = ~ 1 | age),
  data = alc,
  method = "REML"
)

m6 <- lme( # random intercept + heteroske ind errors (gender) model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  weights = varIdent(form = ~ 1 | male),
  data = alc,
  method = "REML"
)

cov_table <- tibble(
  `Covariance model` = c(
    "Random intercepts + independent errors",
    "Random intercepts and slopes + independent errors",
    "Random intercepts + AR(1) errors",
    "Random intercepts + exponential errors",
    "Random intercepts + heteroskedastic independent errors by age",
    "Random intercepts + heteroskedastic independent errors by gender"
  ),
  AIC = c(AIC(m1), AIC(m2), AIC(m3), AIC(m4), AIC(m5), AIC(m6)),
  BIC = c(BIC(m1), BIC(m2), BIC(m3), BIC(m4), BIC(m5), BIC(m6))
) %>%
  mutate(
    AIC = round(AIC, 3),
    BIC = round(BIC, 3)
  )

paper_kable(cov_table, "Comparison of covariance structures")
```

Table 3: Comparison of covariance structures

Covariance model	AIC	BIC
Random intercepts + independent errors	650.930	712.816
Random intercepts and slopes + independent errors	637.832	706.594
Random intercepts + AR(1) errors	644.765	710.089
Random intercepts + exponential errors	644.765	710.089
Random intercepts + heteroskedastic independent errors by age	647.406	716.168
Random intercepts + heteroskedastic independent errors by gender	652.107	717.431

Random intercept and slope model is the best.

```
m_mean1 <- lme( # mean effect model
  alcuse ~ age + coa + peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean2 <- lme( # mean effect model with gender
  alcuse ~ age + coa + peer + male,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean3 <- lme( # two way interaction model
  alcuse ~ age + coa + peer + age:coa + age:peer + coa:peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean4 <- lme( # two way interaction model with gender
  alcuse ~ age + coa + peer + male + age:coa + age:peer + coa:peer + male:age + male:peer + male:coa,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean5 <- lme( # three way interaction model
  alcuse ~ age * coa * peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean6 <- lme( # three way interaction model with gender
  alcuse ~ age * coa * peer * male,
  random = ~ age | id,
  data = alc,
  method = "ML"
```

```
)
```

```r
# without gender model comparison using anova()
mean_model_compare <- anova(m_mean1, m_mean3, m_mean5) %>%
  as.data.frame() %>%
  mutate(
    Model = c(
      "Main effects",
      "Main effects + two-way interactions",
      "Main effects + two-way + three-way interaction"
    )
  ) %>%
  select(Model, df, logLik, `L.Ratio`, `p-value`) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

rownames(mean_model_compare) <- NULL

# writing the model output into a table
paper_kable(
  mean_model_compare,
  "Likelihood ratio tests for fixed-effects structure selection without gender"
)
```

Table 4: Likelihood ratio tests for fixed-effects structure selection without gender

Model	df	logLik	L.Ratio	p-value
Main effects	8	-295.925	NA	NA
Main effects + two-way interactions	11	-291.296	9.259	0.026
Main effects + two-way + three-way interaction	12	-289.038	4.516	0.034

For without gender, three way interaction is the best.

```r
# with gender model comprison using anova()
mean_model_compare <- anova(m_mean2, m_mean4, m_mean6) %>%
  as.data.frame() %>%
  mutate(
    Model = c(
      "Main effects",
      "Main effects + two-way interactions",
      "Main effects + two-way + three-way interaction"
    )
  ) %>%
  select(Model, df, logLik, `L.Ratio`, `p-value`) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

rownames(mean_model_compare) <- NULL

# writing the model output into a table
paper_kable(
  mean_model_compare,
  "Likelihood ratio tests for fixed-effects structure selection with gender"
)
```

Table 5: Likelihood ratio tests for fixed-effects structure selection with gender

Model	df	logLik	L.Ratio	p-value
Main effects	9	-295.347	NA	NA
Main effects + two-way interactions	15	-287.338	16.018	0.014
Main effects + two-way + three-way interaction	20	-283.782	7.111	0.213

For with gender, two way interaction is the best.

```
# AIC and BIC to compare the two models
mean_table <- tibble(
  `Model` = c(
    "Two-way interactions + gender",
    "Three-way interaction"
  ),
  AIC = c(
    AIC(m_mean4), AIC(m_mean5)
  ),
  BIC = c(
    BIC(m_mean4), BIC(m_mean5)
  )
) %>%
  mutate(
    AIC = round(AIC, 3),
    BIC = round(BIC, 3)
  )

paper_kable(mean_table, "Model Comparison")
```

Table 6: Model Comparison

Model	AIC	BIC
Two-way interactions + gender	604.675	657.255
Three-way interaction	602.075	644.139

Three way interaction without gender is the best.

Final model

```
final_model <- lme( # model without controlling for gender
  alcuse ~ age * coa * peer,
  random = ~ age | id,
  data = alc,
  method = "REML"
)


# print the result of the chosen model: three way interaction model
```

```r
coef_table <- summary(final_model)$tTable %>%
  as.data.frame() %>%
  rownames_to_column("Variable") %>%
  rename(
    Estimate = Value,
    SE = Std.Error,
    DF = DF,
    t = `t-value`,
    `p-value` = `p-value`
  ) %>%
  mutate(
    across(where(is.numeric), ~ round(.x, 3))
  ) %>%
  mutate(
    Variable = dplyr::recode(
      Variable,
      "(Intercept)" = "Constant",
      "age" = "Age",
      "coa1" = "Parental alcoholism",
      "peer" = "Peer alcohol use",
      "age:coa1" = "Age × parental alcoholism",
      "age:peer" = "Age × peer alcohol use",
      "coa1:peer" = "Parental alcoholism × peer alcohol use",
      "age:coa1:peer" = "Age × parental alcoholism × peer alcohol use"
    )
  )

paper_kable(
  coef_table,
  "Fixed effects estimates for the final mixed-effects model"
)
```

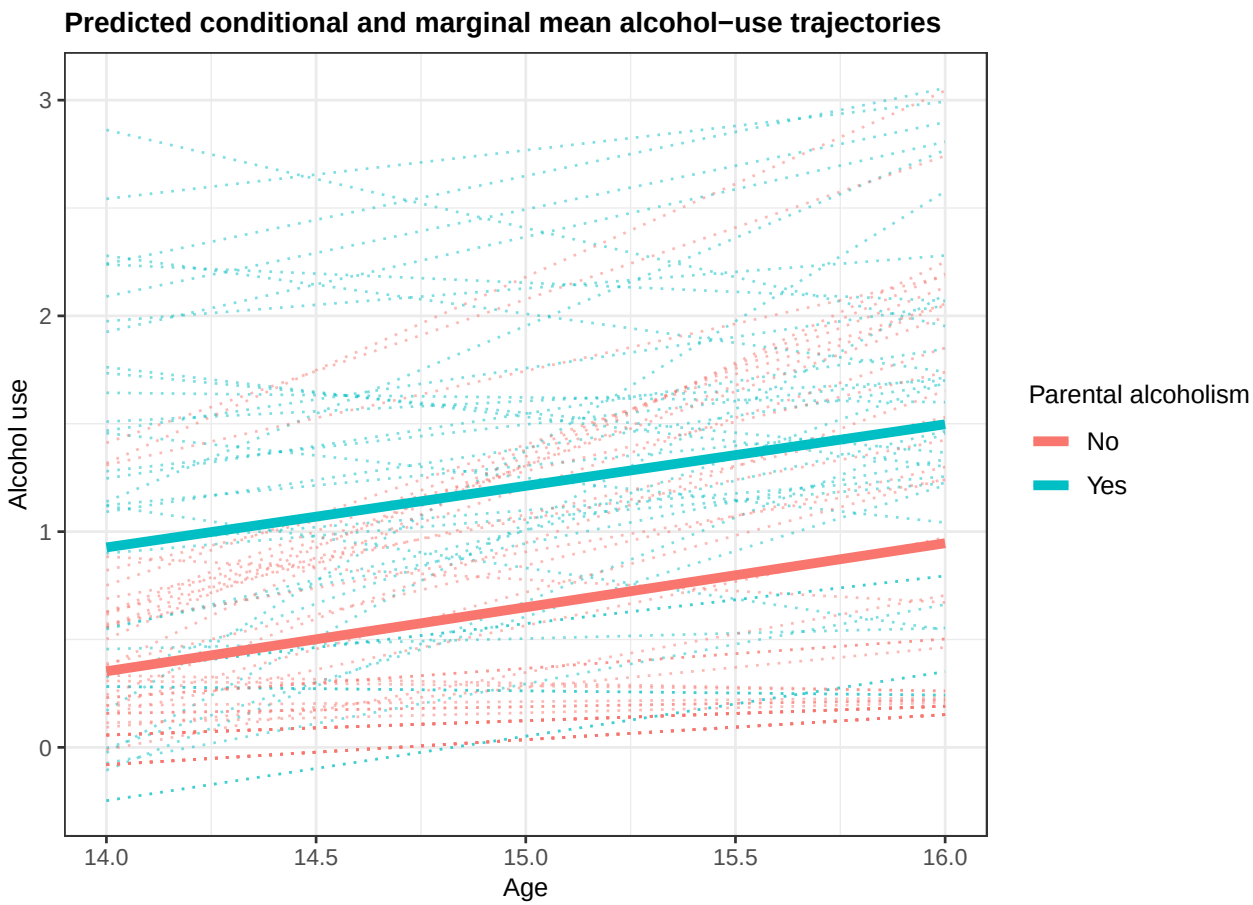
Table 7: Fixed effects estimates for the final mixed-effects model

Variable	Estimate	SE	DF	t	p-value
Constant	-3.639	2.022	160	-1.800	0.074
Age	0.260	0.139	160	1.870	0.063
Parental alcoholism	-5.074	3.125	78	-1.624	0.108
Peer alcohol use	-0.160	1.788	78	-0.090	0.929
Age × parental alcoholism	0.354	0.215	160	1.648	0.101
Age × peer alcohol use	0.036	0.123	160	0.293	0.770
Parental alcoholism × peer alcohol use	5.707	2.486	78	2.296	0.024
Age × parental alcoholism × peer alcohol use	-0.359	0.171	160	-2.102	0.037

Prediction plot

```r
# compute conditional predicted values from the final model
condmean <- predict(final_model, level = 1)
```

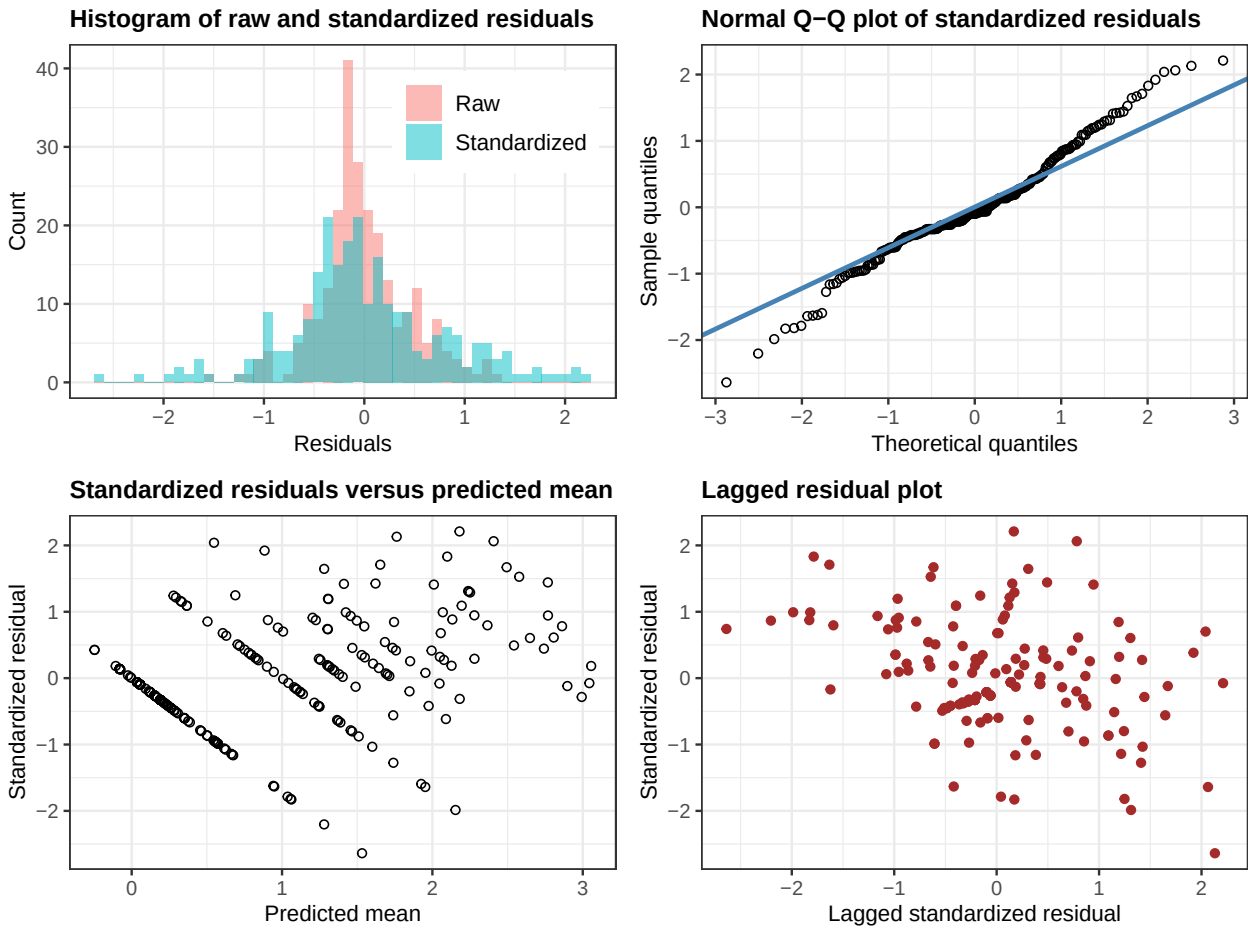
```r
# compute the mean peer alcohol use
peer0 <- mean(alc$peer, na.rm = TRUE)

# create new data for marginal predictions across age and coa
margdat <- expand.grid(
  age = sort(unique(alc$age)),
  coa = levels(alc$coa),
  peer = peer0
)

# compute marginal predicted mean alcohol use
margdat$margmean <- predict(final_model, newdata = margdat, level = 0)

# combine predicted values with the original dataset
predalc <- as.data.frame(cbind(alc, condmean))

# plot of predicted alcohol use trajectories
ggplot(
  predalc,
  aes(x = age, y = condmean, group = id)
) +
  geom_line(aes(color = coa), linetype = 3, alpha = 0.5) +
  geom_line(
    data = margdat,
    aes(x = age, y = margmean, color = coa, group = coa),
    linewidth = 1.8,
    inherit.aes = FALSE
  ) +
  scale_color_manual(values = coa_cols, labels = c("No", "Yes"),
                     name = "Parental alcoholism") +
  labs(
    title = "Predicted conditional and marginal mean alcohol-use trajectories",
    x = "Age",
    y = "Alcohol use"
  ) +
  base_plot_theme
```



Model diagnostic

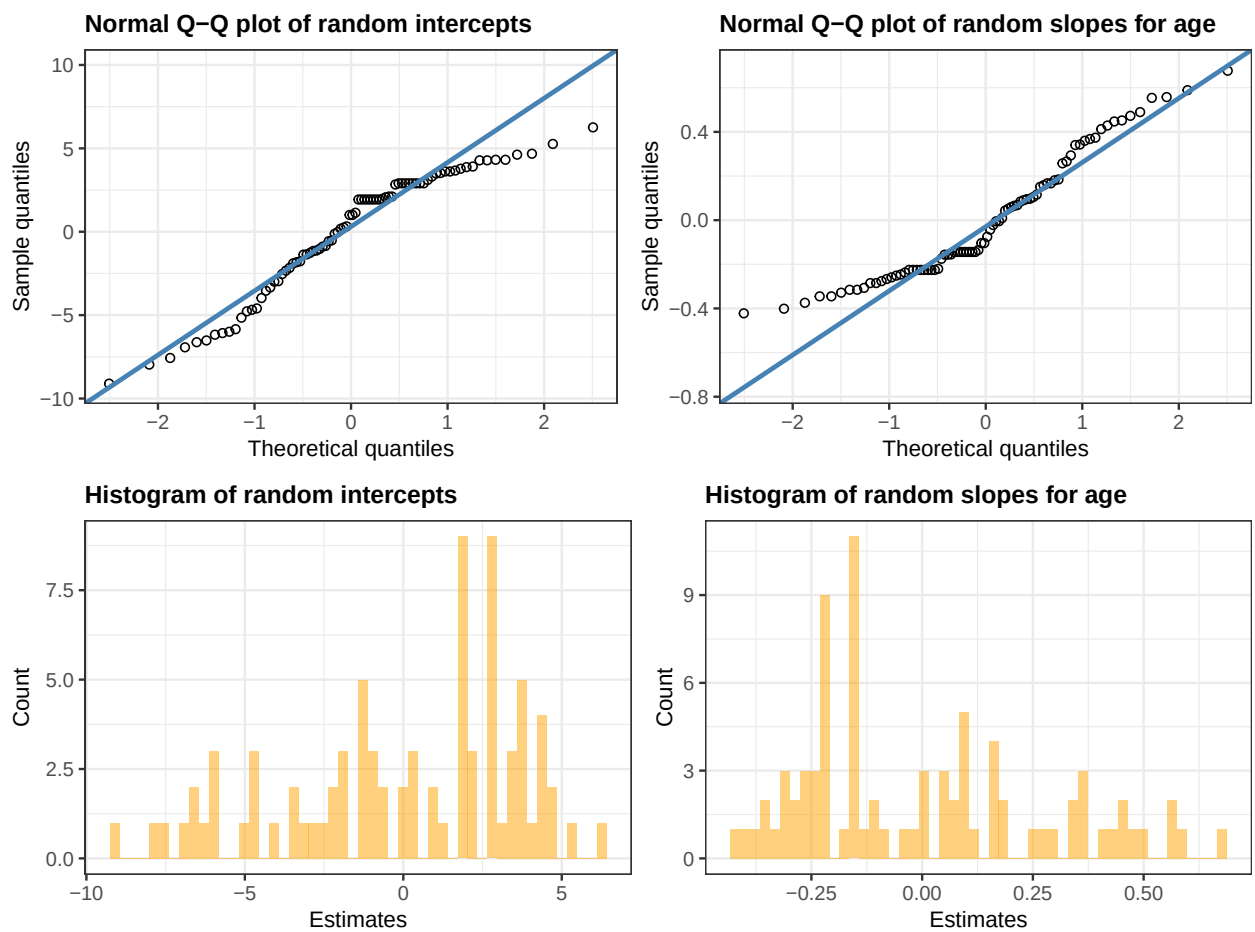
```
rhat <- resid(final_model, type = "normalized")

condmean <- predict(final_model)
y <- alc$alcuse
stg1r <- y - condmean

rtype <- c(rep("Raw", length(stg1r)), rep("Standardized", length(stg1r)))
resid <- c(stg1r, rhat)

stg1resids <- data.frame(
  rtype = rtype,
  resid = as.numeric(resid)
)

p1 <- ggplot(stg1resids, aes(x = resid, fill = rtype)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50) +
  labs(
    title = "Histogram of raw and standardized residuals",
    y = "Count",
    x = "Residuals",
    fill = NULL
```

```r
  ) +
  theme(
    legend.position = c(0.78, 0.78)
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(rhat = rhat), aes(sample = rhat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of standardized residuals",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

yhat <- predict(final_model)

p3 <- ggplot(data.frame(yhat = yhat, rhat = rhat), aes(y = rhat, x = yhat)) +
  geom_point(shape = 1) +
  labs(
    title = "Standardized residuals versus predicted mean",
    x = "Predicted mean",
    y = "Standardized residual"
  ) +
  base_plot_theme

p4 <- alc %>%
  mutate(rhat = rhat) %>%
  group_by(id) %>%
  mutate(rhat1 = lag(rhat)) %>%
  ggplot(aes(y = rhat, x = rhat1)) +
  geom_point(color = "brown", na.rm = TRUE) +
  labs(
    title = "Lagged residual plot",
    y = "Standardized residual",
    x = "Lagged standardized residual"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```




```
re <- ranef(final_model) %>%
  rownames_to_column("id")

b0hat <- re$`(Intercept)`
b1hat <- re$age

p1 <- ggplot(data.frame(b0hat = b0hat), aes(sample = b0hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of random intercepts",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(b1hat = b1hat), aes(sample = b1hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of random slopes for age",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
```

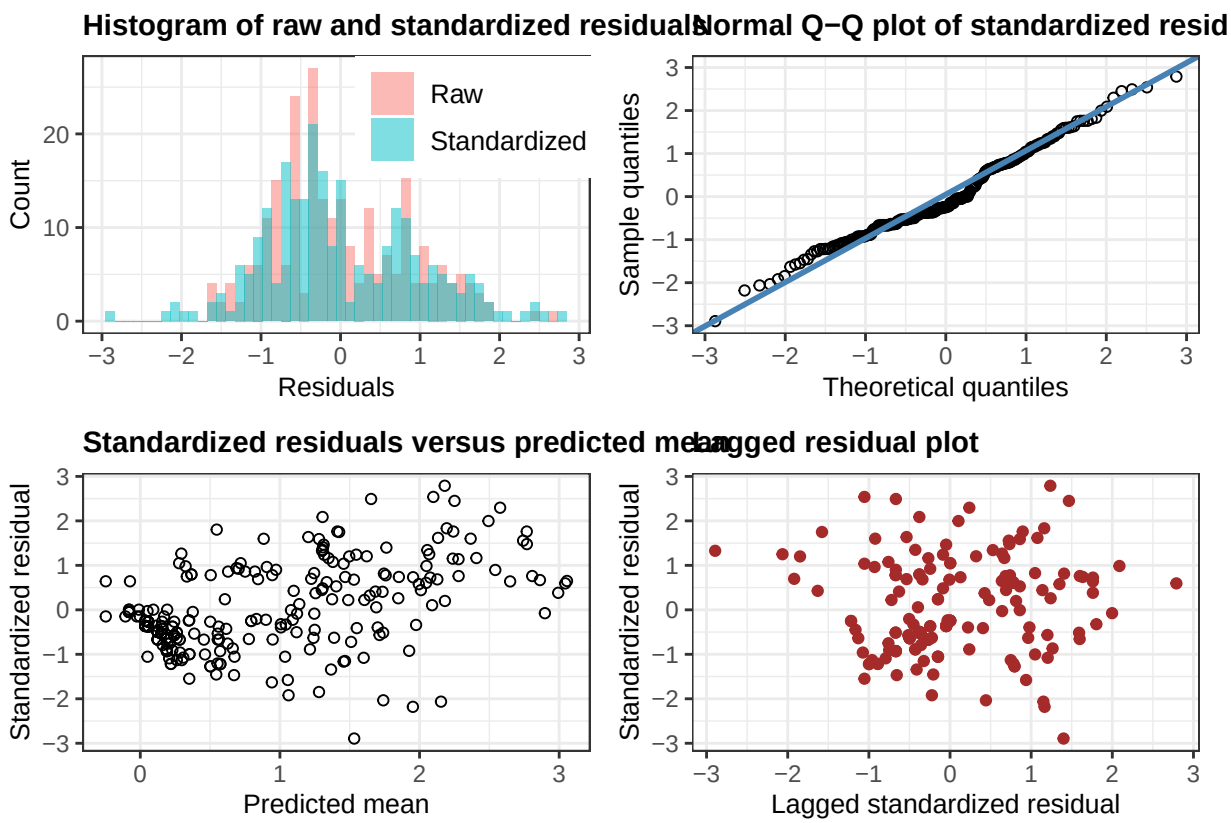
```r
  base_plot_theme

p3 <- ggplot(data.frame(b0hat = b0hat), aes(x = b0hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random intercepts",
    x = "Estimates",
    y = "Count"
  ) +
  base_plot_theme

p4 <- ggplot(data.frame(b1hat = b1hat), aes(x = b1hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random slopes for age",
    x = "Estimates",
    y = "Count"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```



Try the random intercept only model

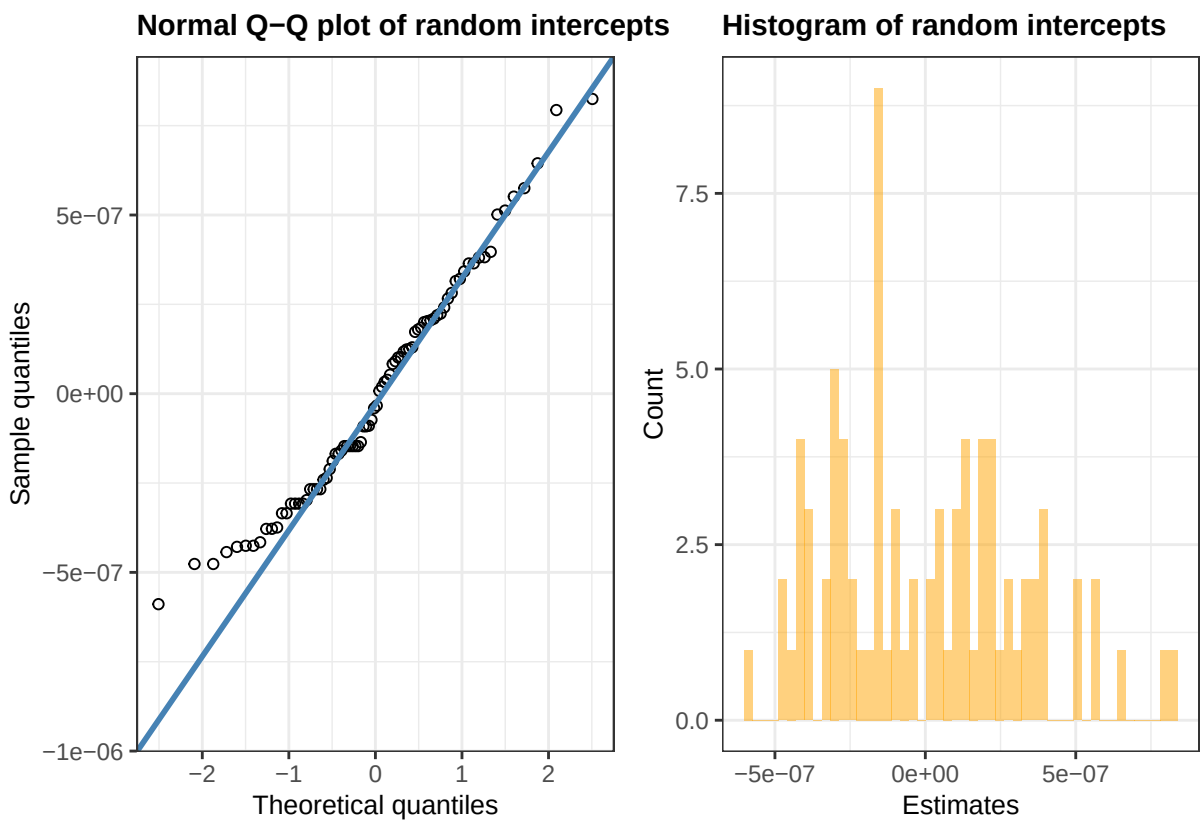
```
m3 <- lme( # random intercept + AR1 errors model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "REML"
)
rhat <- resid(m3, type = "normalized")

condmean <- predict(m3)
y <- alc$alcuse
stg1r <- y - condmean

rtype <- c(rep("Raw", length(stg1r)), rep("Standardized", length(stg1r)))
resid <- c(stg1r, rhat)

stg1resids <- data.frame(
  rtype = rtype,
  resid = as.numeric(resid)
)

p1 <- ggplot(stg1resids, aes(x = resid, fill = rtype)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50) +
  labs(
    title = "Histogram of raw and standardized residuals",
    y = "Count",
    x = "Residuals",
    fill = NULL
  ) +
  theme(
    legend.position = c(0.78, 0.78)
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(rhat = rhat), aes(sample = rhat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of standardized residuals",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

yhat <- predict(final_model)

p3 <- ggplot(data.frame(yhat = yhat, rhat = rhat), aes(y = rhat, x = yhat)) +
  geom_point(shape = 1) +
  labs(
    title = "Standardized residuals versus predicted mean",
    x = "Predicted mean",
```

```
    y = "Standardized residual"
  ) +
  base_plot_theme

p4 <- alc %>%
  mutate(rhat = rhat) %>%
  group_by(id) %>%
  mutate(rhat1 = lag(rhat)) %>%
  ggplot(aes(y = rhat, x = rhat1)) +
  geom_point(color = "brown", na.rm = TRUE) +
  labs(
    title = "Lagged residual plot",
    y = "Standardized residual",
    x = "Lagged standardized residual"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```




```
re <- ranef(m3) %>%
  rownames_to_column("id")

b0hat <- re$`(Intercept)`

p1 <- ggplot(data.frame(b0hat = b0hat), aes(sample = b0hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
```

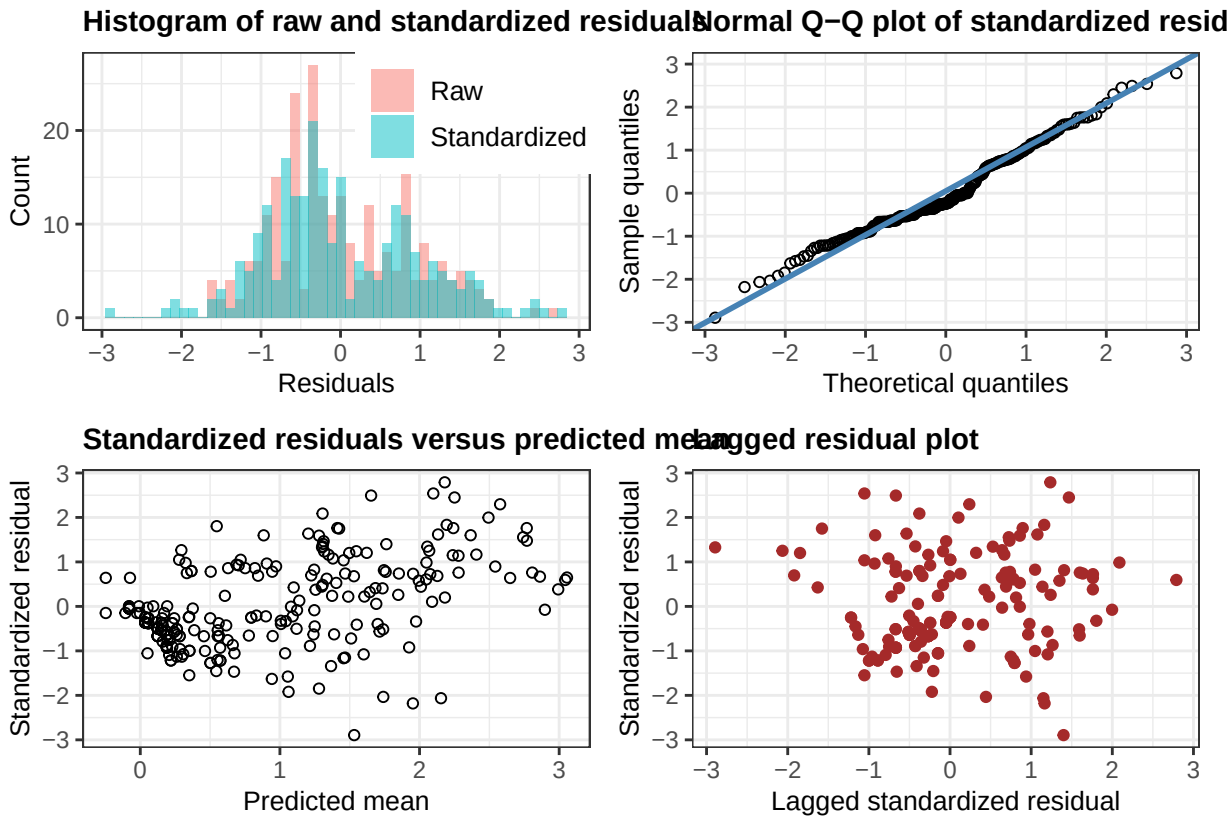
```r
  labs(
    title = "Normal Q-Q plot of random intercepts",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

p3 <- ggplot(data.frame(b0hat = b0hat), aes(x = b0hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random intercepts",
    x = "Estimates",
    y = "Count"
  ) +
  base_plot_theme


(p1 + p3)
```

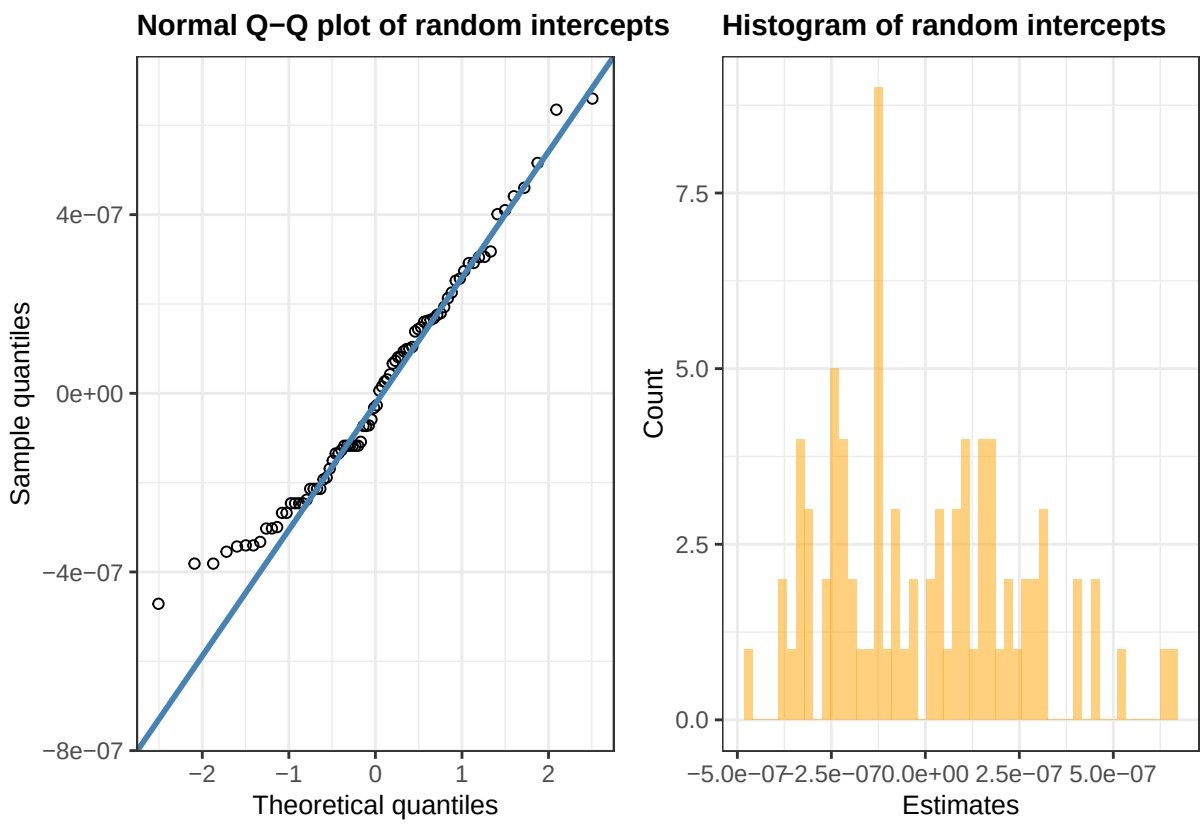



```r
m4 <- lme( # random intercept + exp errors model
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  correlation = corExp(form = ~ age | id),
  data = alc,
  method = "REML"
)
rhat <- resid(m4, type = "normalized")
```

```r
condmean <- predict(m4)
y <- alc$alcuse
stg1r <- y - condmean

rtype <- c(rep("Raw", length(stg1r)), rep("Standardized", length(stg1r)))
resid <- c(stg1r, rhat)

stg1resids <- data.frame(
  rtype = rtype,
  resid = as.numeric(resid)
)
p1 <- ggplot(stg1resids, aes(x = resid, fill = rtype)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50) +
  labs(
    title = "Histogram of raw and standardized residuals",
    y = "Count",
    x = "Residuals",
    fill = NULL
  ) +
  theme(
    legend.position = c(0.78, 0.78)
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(rhat = rhat), aes(sample = rhat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of standardized residuals",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

yhat <- predict(final_model)

p3 <- ggplot(data.frame(yhat = yhat, rhat = rhat), aes(y = rhat, x = yhat)) +
  geom_point(shape = 1) +
  labs(
    title = "Standardized residuals versus predicted mean",
    x = "Predicted mean",
    y = "Standardized residual"
  ) +
  base_plot_theme

p4 <- alc %>%
  mutate(rhat = rhat) %>%
  group_by(id) %>%
  mutate(rhat1 = lag(rhat)) %>%
  ggplot(aes(y = rhat, x = rhat1)) +
  geom_point(color = "brown", na.rm = TRUE) +
  labs(
    title = "Lagged residual plot",
```

```
    y = "Standardized residual",
    x = "Lagged standardized residual"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```




```
re <- ranef(m4) %>%
  rownames_to_column("id")

b0hat <- re$`(Intercept)`

p1 <- ggplot(data.frame(b0hat = b0hat), aes(sample = b0hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of random intercepts",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

p3 <- ggplot(data.frame(b0hat = b0hat), aes(x = b0hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random intercepts",
    x = "Estimates",
```

```
    y = "Count"
  ) +
  base_plot_theme

(p1 + p3)
```